



KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

KnowSelf introduces agentic knowledgeable self-awareness, enabling LLM agents to autonomously decide when to act directly, self-reflect, or consult external knowledge during planning tasks.

KnowSelf modes

Fast • Slow •
Knowledgeable

Selective reflection + retrieval

Current Agent Training Uses 'Flood Irrigation' — Indiscriminate Knowledge Injection Hurts



Trajectory imitation (e.g., ReAct)

- Blindly imitates trajectories regardless of need
- Fragile when inputs deviate from patterns



Trial-and-error refinement (e.g., Reflexion)

- Injects external feedback broadly (not situational)
- Extra inference cost even when unnecessary



Humans use situational self-awareness: act directly vs. reflect vs. seek external help.

(“100% Know%”)

arxiv.org/abs/2504.03553

High cost; can hurt if knowledge is irrelevant

“Flood irrigation” knowledge injection



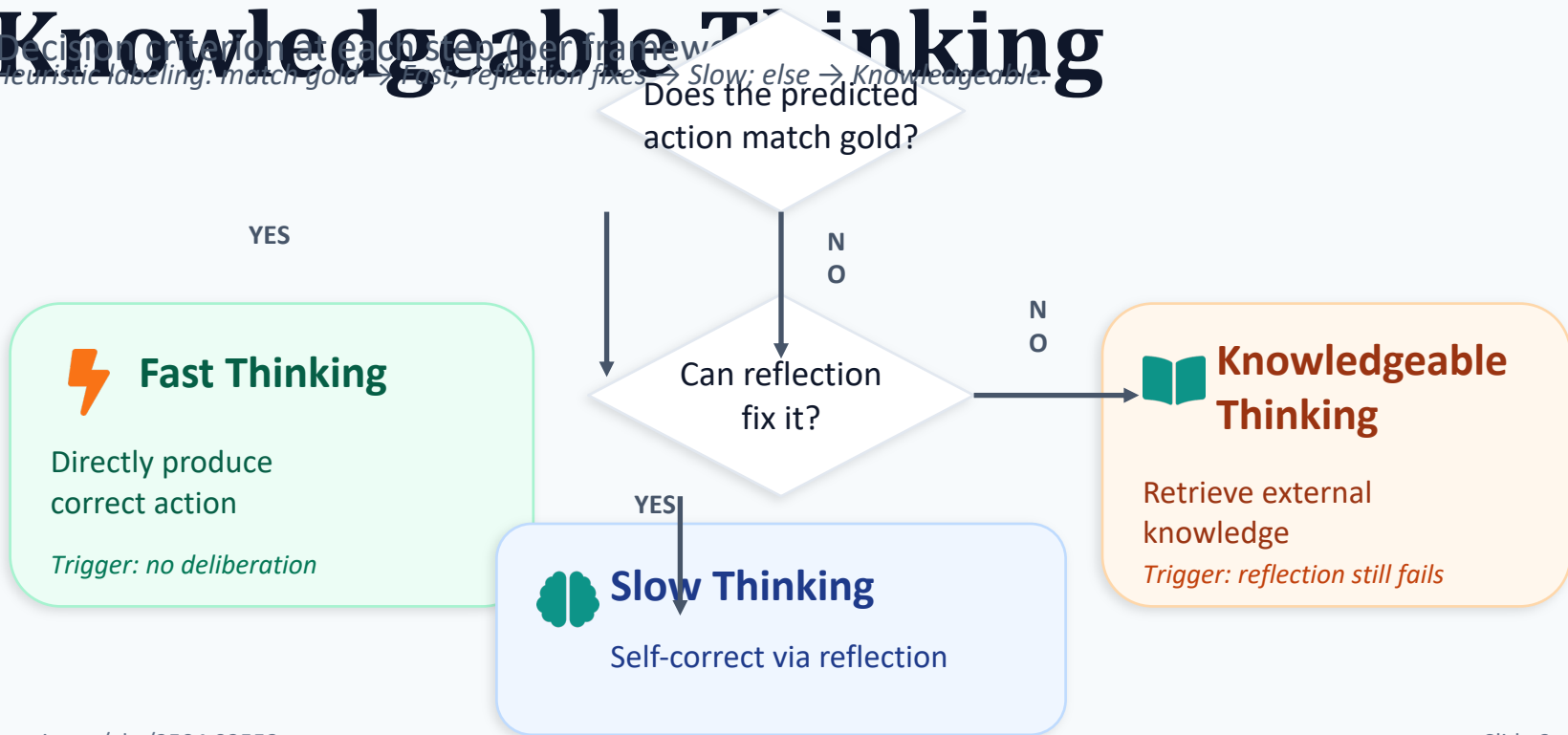
Gold trajectories / feedback / knowledge

Unnecessary cost

Fragile planning

Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable Thinking

Decision criterion at each step (per framework):
Heuristic labeling: match gold $\rightarrow$ Fast; reflection fixes $\rightarrow$ Slow; else $\rightarrow$ Knowledgeable.



KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference

1

Self-awareness data construction

Self-explore environments, label each step via heuristic judgment

- Insert special tokens marking Fast / Slow / Knowledgeable situation

2

Two-stage training

- SFT for initial self-awareness patterns
- RPO to refine special-token generation

3

Self-aware inference

- Generate tokens at runtime to trigger reflection or retrieval

- Avoid knowledge queries unless needed

Framework overview (Fig. 3)

Step 1

Self-awareness Data Construction

Task: put a clean egg in microwave

Action: go to fridge 1
Observation: The fridge 1 is closed.
Action: open fridge 1
Observation: The fridge 1 is open. In it, you see a cup 3, a cup 1, a lettuce 1, an egg 2.

Action: close fridge 1
Action: take egg 2 from fridge 1

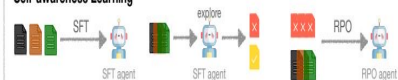
Reflection: <> need a clean egg, but egg 2 is not clean. I need to search for another place to find a clean egg.</>
Action: go to drawer 1

Reflection: <> Oh, it seems that I missed the egg. I need to pick it up and clean it now.</>
Action: take egg 2 from fridge 1

Knowledge: <> To obtain a cleaned objective, you should first find the objective and then clean it, rather than search for a cleaned objective directly.</>
Action: take egg 2 from fridge 1

Step 2

Self-awareness Learning



Step 3

Self-awareness Inference

Task: put two newspapers in drawer

Action: go to sofa 1
Observation: On the sofa 1, you see a creditcard 2, a newspaper 1.
Action: take newspaper 1 from sofa 1
Observation: You pick up the newspaper 1 from the sofa 1.
Action: go to counter 2
Observation: On the counter 2, you see a newspaper 2.

Fast Thinking
Action: take newspaper 2 from counter 2

Reflection: <> Wait, let's check the action. Sorry, I made a mistake. I need put the first newspaper before picking another one.</>
Action: go to drawer 2

Knowledgeable Thinking
Knowledge: <> When the agent has taken a target object and needs to take additional items to fulfill a task, the agent should first place the current object at its target receptacle, ensuring it is holding nothing.</>
Action: go to drawer 2

NEXT STEP

KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

Qualitative evidence (Fig. 1)

Example A: “put a hot mug in cabinet”

- OpenAI-O1: opens microwave, then mis-targets an egg
- KnowSelf (Llama-8B): uses Reflection to heat the mug using the open microwave

Example B: “put a saltshaker in drawer”

- DeepSeek-R1: reflects but gets stuck on closing drawer
- KnowSelf: applies Knowledge token—place object without removing unrelated items

Figure 1 (cropped):



KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only

Success rate (%) across ALFWorld task types; Know% = fraction of steps retrieving external knowledge

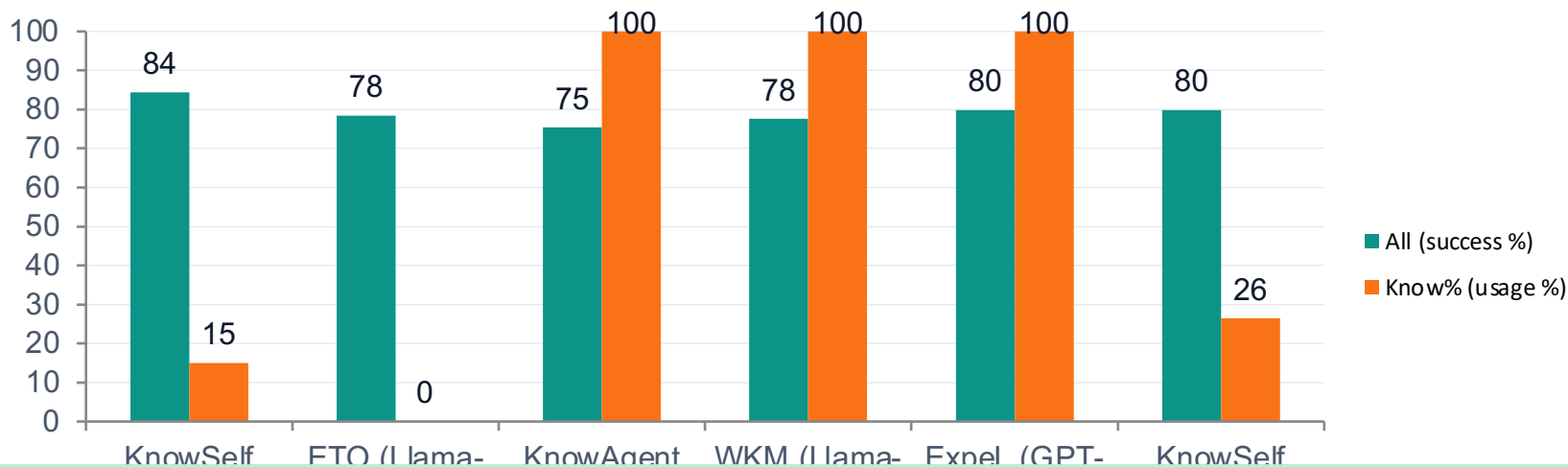
Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37

KnowSelf (Llama-8B): 84.33% All with 15.01% Know% • KnowSelf (Gemma-2B): 79.85% All with 26.41% Know%

Gemma-2B	ReAct	0%	0.00	9.09	0.00	4.76	44.44	0.00	8.90
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	81.67	83.33	82.57	71.43	61.67	41.40	73.20

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage Over Full-Knowledge

Know% (knowledge usage) vs overall success (All)



Efficiency: KnowSelf uses ~15–26% knowledge vs 100% for Expel/KnowAgent/WKM, yet reaches best (or tied-best) All.



Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

1. Indiscriminate knowledge injection (“flood irrigation”) is wasteful and can be suboptimal.
2. Situational self-awareness can be learned via special tokens with heuristic labeling on self-explored trajectories.
3. KnowSelf (Llama-8B) exceeds GPT-4o ExpeL on ALFWorld: 84.33% vs 79.85% All.
4. Selective knowledge is enough: 15.01–26.41% Know% vs 100% for full-knowledge methods.

5. Scales across model sizes: Gemma-2B KnowSelf reaches 79.85% (matching GPT-4o ExpeL).

Future: extend situational self-awareness to broader tools and environments; refine judgment signals beyond heuristics.